

DATA ANALYSIS FOR TRACKING AND OPTIMIZING BOILER PERFORMANCE

The [original paper](#) contains 29 sections, with 10 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Introduction

By nature, any data point collected will contain a certain level of uncertainty due to systematic and random errors. In power plants, fuel and environmental conditions are seldom steady. As a first step in the analysis process, the collected data is thoroughly reviewed for any inconsistencies and/ or errors.

SUMMARY PASSAGE 2

Instrument Calibration:

The graph in Figure 1 shows the temperature to range from 600 -970°F (316 -521°C), giving a total average 829°F (443°C). To assess the validity of the thermocouple data, the DCS data for the bulk average high temperature superheater outlet temperature was used and it shows the steam temperature to be 950°F (510°C). This outlet temperature of 950°F (510°C) aligns with the predicted design conditions at the current operating load.

SUMMARY PASSAGE 3

Data Validation Via Manual Calculation

Certain parameters for a boiler will be calculated and verified by performing a hand calculation. Often, measured spray flow will be verified by performing a heat balance calculation and the verified flow will be used to determine main steam flow.

SUMMARY PASSAGE 4

Figure 3: Spray Flow Diagram

Using uncertainty analysis as described by ASME PTC 19.1, it can be determined that the total uncertainty of the calculated result is approximately 18%. The uncertainty of the measured spray flow is approximately 20% assuming an uncalibrated and uninspected flow nozzle and a standard deviation of the measurement of 10% of the mean. It is often noted that although other major operating parameters (e.g. steam flow, temperature and pressure) may meet the criteria for operational stability, the measured spray flow often has a high standard deviation which leads to a high random uncertainty being associated with the measurement.

SUMMARY PASSAGE 5

Data Collection

The more data points collected, the better the representation of the boiler performance will be. Having additional data points reduces the overall random uncertainty of the data points collected. ASME PTC 4-2013 section 4-3.3 specifies that data (other than quantity measurements, which are based on the device) should preferably be collected at an interval of two minutes or less [1] for testing.

SUMMARY PASSAGE 6

Figure 5: Data Frequency Example

As an example, Figure 5 represents 120 data points giving boiler feedwater flow collected at one minute increments. This data was then narrowed down to data points representing collection at five and fifteen minute increments. It is evident that as the frequency of the data is reduced (from minute by minute points to every 15 minute points), the general trend of all three time increments is the same, since they all pass through the same data value points.

SUMMARY PASSAGE 7

Predicting Data Values:

Since FEGT at 100% MCR needs to be predicted. As a first step, the known values for FEGT (y-axis) and %MCR (y-axis) at the specified boiler loads are plotted against their corresponding HRR (x-axis) as shown in Figure 6 and Figure 7 respectively. Using the trend developed for %MCR vs. HRR (Figure 7), the HRR at 100% MCR was calculated.

SUMMARY PASSAGE 8

Figure 7: Steam Flow Vs. Heat Release Rate

This caused the steam temperature to drop below the lower range of its original design and caused the Economizer outlet gas temperature to drop. To close out this discussion on predicting data values, note that for a typical boiler, the FEGT vs. HRR does not have a linear relationship. In order to increase the accuracy of the relationship between FEGT to HRR, at minimum, three data sets are required.

SUMMARY PASSAGE 9

Data Representation:

The sample size being analyzed can be extremely important, not only for reducing uncertainty during a steadystate test, but also for characterizing unit operation for a design basis in an upgrade project. Figure 8 shows superheat steam temperature plotted for a complete year of operation at or near full load operation along with several test results. The plot shows that for similar steam flows, the superheat temperature varied by 50°F (122°C) or more throughout the sampled time frame.

SUMMARY PASSAGE 10

Figure 8: Long-Term Trends And Test Data

In this example, further analysis of this particular boiler at full load revealed that fouling of the superheater surface results in depressed temperatures at the higher loads. When the superheater surfaces are cleaned, steam temperatures return to their design values. The project needs to properly account for the variation in the cleanliness of the superheater surface.